

Project - Titanic Passenger Survival Analysis

Problem Statement

- Perform Exploratory Data Analysis(EDA) using matplotlib library
- Create a insightful visualization to understand the factors that infulenced survival on the Titanic

```
In [60]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats
%matplotlib inline
```

Dataset

- Dataset **Titanic-Dataset.csv** is downloaded from Kaggle
- Link: <https://www.kaggle.com/datasets/yasserh/titanic-dataset>
- This dataset contains details of all passengers when the titanic drowned and some people died
- This dataset may contains of missing values

Loading dataset

- Reading the dataset using **read_csv** method from pandas
- This function converts the dataset in **Pandas DataFrame**

```
In [61]: df = pd.read_csv('Titanic-Dataset.csv')
df.head()
```

Out [61]:	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Data Preparation

- Check null values using **isna()** method
- Check unique values using **unique()** method
- **sum()** displays how many values are null in particular column

```
In [62]: df.isna().sum()
```

```
Out [62]: PassengerId    0
Survived    0
Pclass      0
Name        0
Sex         0
Age        177
SibSp       0
Parch       0
Ticket      0
Fare        0
Cabin     687
Embarked    2
dtype: int64
```

- Now we have to check manually which are mandatory columns that shouldn't have null values
- As per the problem statement we need:
 1. Survived
 2. Pclass
 3. Sex
 4. Age
 5. Fare
- The **Age** column has some null values let's handle them

```
In [63]: df['Age'] = df['Age'].fillna(df['Age'].mean())
```

```
In [64]: # again check null values for confirmation
df.isna().sum()
```

```
Out [64]: PassengerId      0
Survived      0
Pclass        0
Name          0
Sex           0
Age           0
SibSp         0
Parch         0
Ticket        0
Fare          0
Cabin        687
Embarked      2
dtype: int64
```

EDA tasks

1. Overall survival rate
2. Survival rate based on gender
3. Survival rate based on passenger class
4. Survival rate based on age group
5. Age distribution
6. Fare vs age

Caluculations for visualization

Overall survival rate

```
In [65]: # Overall survival rate
survived = (df['Survived']==1).sum()
died = (df['Survived']==0).sum()
total = df['Survived'].count()
print(survived,died,total)
```

```
342 549 891
```

Survival rate based on gender

```
In [66]: # Survival rate based on gender - 'Sex'
gender_survival = df.groupby('Sex')['Survived'].mean()*100
gender_survival
```

```
Out [66]: Sex
female    74.203822
male      18.890815
Name: Survived, dtype: float64
```

```
In [67]: # additional information based on gender - 'Sex'
gender_survived = df.groupby('Sex')['Survived'].sum()
gender_died = df.groupby('Sex')['Survived'].count() - df.groupby('Sex')['Survived'].sum()
gender_total = df.groupby('Sex')['Survived'].count()

print(gender_survived,gender_died,gender_total)
```

```
Sex
female    233
male       109
Name: Survived, dtype: int64
Sex
female      81
male       468
```

```
Name: Survived, dtype: int64 Sex
female    314
male      577
Name: Survived, dtype: int64
```

Survival rate based on passenger class

```
In [68]: # Survival rate based on passenger class - 'Pclass'
Pclass_survival = df.groupby('Pclass')['Survived'].mean()*100

print(Pclass_survival)
```

```
Pclass
1    62.962963
2    47.282609
3    24.236253
Name: Survived, dtype: float64
```

```
In [69]: # additional information based on passenger class - 'Pclass'
Pclass_total = df.groupby('Pclass')['Survived'].count()
Pclass_survived = df.groupby('Pclass')['Survived'].sum()
Pclass_died = Pclass_total - df.groupby('Pclass')['Survived'].sum()

print(Pclass_survived,Pclass_died,Pclass_total)
```

```
Pclass
1    136
2     87
3    119
Name: Survived, dtype: int64 Pclass
1     80
2     97
3    372
Name: Survived, dtype: int64 Pclass
1    216
2    184
3    491
Name: Survived, dtype: int64
```

Survival rate based on age group

- There is no age group column in the dataset but we have **Age** column
- So based on **Age** we have to create new column named **AgeGroup**
- **AgeGroup** column should contain only these values:
 1. Child
 2. Teen
 3. Adult
 4. Middle Age
 5. Senior

```
In [70]: # Adding 'AgeGroup' column to dataframe
df['AgeGroup'] = pd.cut(df['Age'],bins=[0,12,19,30,45,90],labels=['Child','Teen','Adult','Middle Age']
df.head()
```

```
Out [70]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	AgeGroup
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	Adult
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	Middle Age
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	Adult
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S	Middle Age

PassengerId		Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	AgeGroup
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S	Middle Age

```
In [71]: # Survival rate based on age group - 'AgeGroup'
age_group_survival = df.groupby('AgeGroup')['Survived'].mean()*100

print(age_group_survival)
```

```
AgeGroup
Child      57.971014
Teen       41.052632
Adult      32.938389
Middle Age  42.574257
Senior     36.893204
Name: Survived, dtype: float64
```

C:\Users\user\AppData\Local\Temp\ipykernel_3652\2620449427.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
age_group_survival = df.groupby('AgeGroup')['Survived'].mean()*100
```

```
In [72]: # Additional information based on age group - 'AgeGroup'
age_group_total = df.groupby('AgeGroup')['Survived'].count()
age_group_survived = df.groupby('AgeGroup')['Survived'].sum()
age_group_died = age_group_total - df.groupby('AgeGroup')['Survived'].sum()

print(age_group_survived, age_group_died, age_group_total)
```

```
AgeGroup
Child      40
Teen       39
Adult     139
Middle Age  86
Senior     38
Name: Survived, dtype: int64 AgeGroup
Child      29
Teen       56
Adult     283
Middle Age 116
Senior     65
Name: Survived, dtype: int64 AgeGroup
Child      69
Teen       95
Adult     422
Middle Age 202
Senior    103
Name: Survived, dtype: int64
```

C:\Users\user\AppData\Local\Temp\ipykernel_3652\2742365243.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
age_group_total = df.groupby('AgeGroup')['Survived'].count()
```

C:\Users\user\AppData\Local\Temp\ipykernel_3652\2742365243.py:3: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
age_group_survived = df.groupby('AgeGroup')['Survived'].sum()
```

C:\Users\user\AppData\Local\Temp\ipykernel_3652\2742365243.py:4: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
age_group_died = age_group_total - df.groupby('AgeGroup')['Survived'].sum()
```

Age Distribution

- Age distribution can be done only using **Age**
- We can use **Histogram** for age distribution
- And making more bins helps for better visualization

```
In [73]: # No calculation required for Age Distribution
```

Fare vs Age

- We have to visualize **Fare** and **Age** using **Scatter plot**
- And also we can show who are survived at end based on **Survived** column

```
In [74]: # No calculations for Fare vs Age
```

Data Visualization - Graphical Representation

- Using **subplots** for some important charts
- Additional charts are providing some more information
- Additional charts are represented at last

Core Visualization

```
In [75]: # Defining subplots figure 1
fig1, ax = plt.subplots(2,2,figsize=[10,10])
```

```
In [76]: # Overall survival rate
ax[0,0].pie([survived,died],
            autopct='%0.1f%%',
            labels=['Survived','Died'],
            colors=['green','red'],
            radius=0.8)
ax[0,0].set_title('Overall Survival Rate')

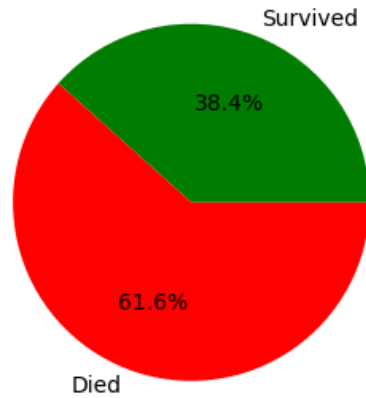
# Survival rate based on gender
xpos1 = np.arange(len(gender_survival))
ax[0,1].bar(xpos1,gender_survival.values,label=gender_survival.index,width=0.6,color='orange')
ax[0,1].set_title('Survival rate based on gender')
ax[0,1].set_xlabel('Gender')
ax[0,1].set_xticks(xpos1,gender_survival.index)
ax[0,1].set_ylabel('Survival rate')

# Survival rate based on passenger class
xpos2 = np.arange(len(Pclass_survival))
ax[1,0].bar(xpos2,Pclass_survival.values,label=Pclass_survival.index,width=0.6,color='green')
ax[1,0].set_title('Survival rate based on Pclass')
ax[1,0].set_xlabel('Pclass')
ax[1,0].set_xticks(xpos2,Pclass_survival.index)
ax[1,0].set_ylabel('Survival rate')

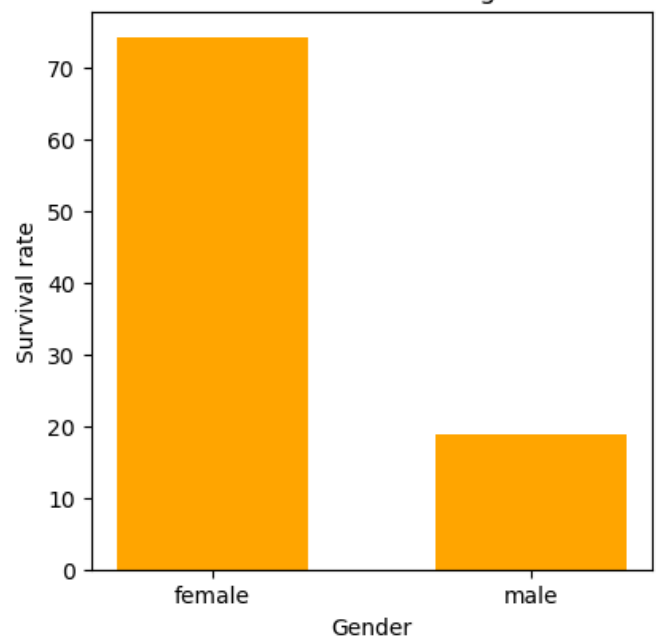
# Survival rate based on age group
xpos3 = np.arange(len(age_group_survival))
ax[1,1].bar(xpos3,age_group_survival.values,label=age_group_survival.index,color='purple')
ax[1,1].set_title('Survival rate based on Age group')
ax[1,1].set_xlabel('Age Group')
ax[1,1].set_xticks(xpos3,age_group_survival.index)
ax[1,1].set_ylabel('Survival rate')

plt.show()
fig1.savefig("Survival_Rate.png", dpi=300)
```

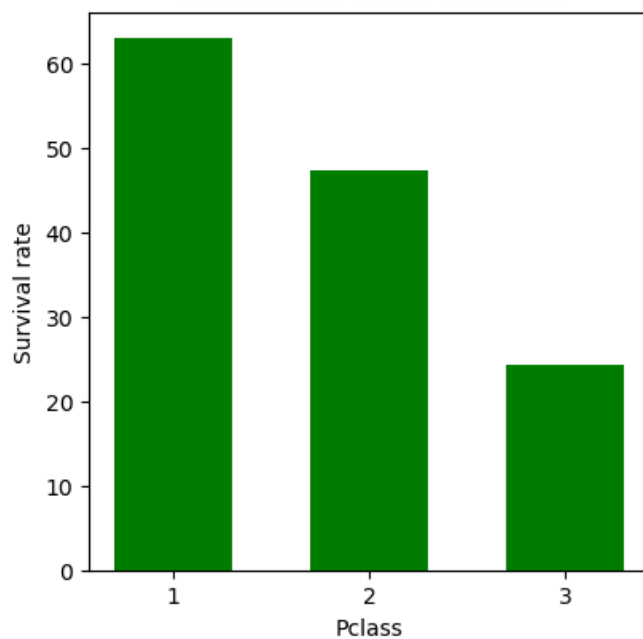
Overall Survival Rate



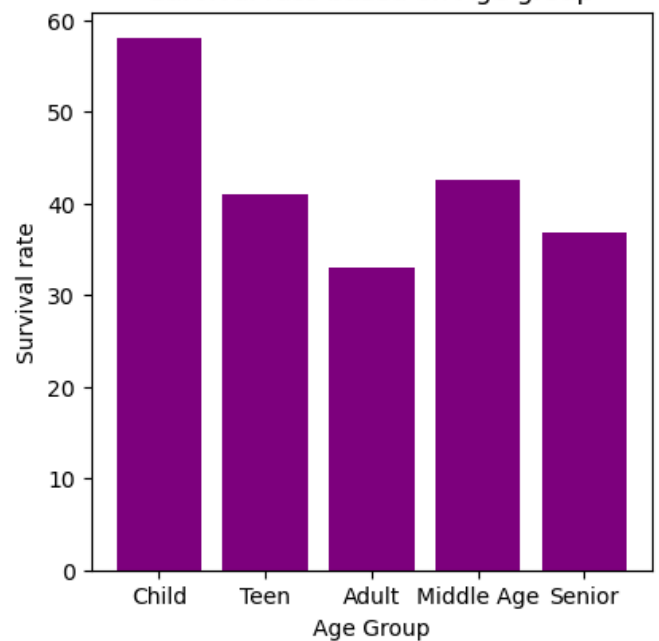
Survival rate based on gender



Survival rate based on Pclass



Survival rate based on Age group



- **INSIGHTS:**

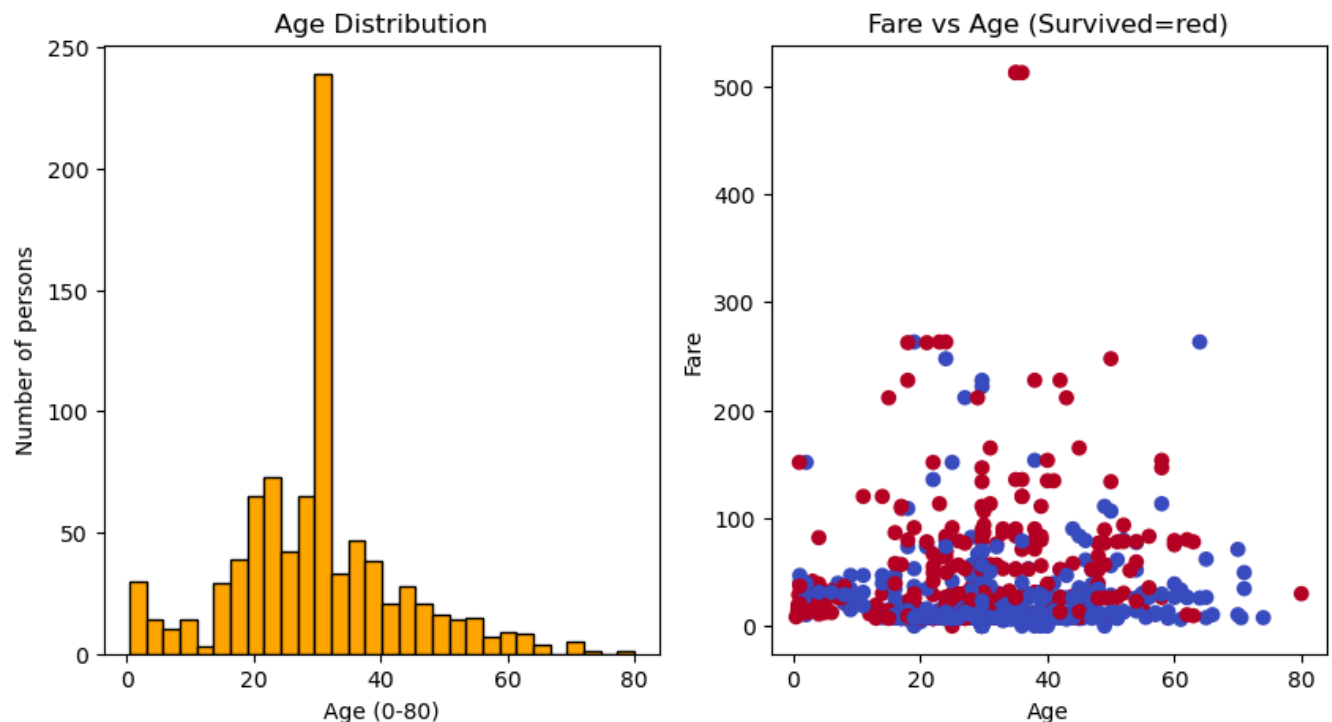
1. **Overall Survival rate :** from total passengers
 - 38.4% i.e 342 were survived
 - 61.6% i.e 549 were died
2. **Survival rate based on gender:**
 - from total female passengers 314 only 233 i.e 74.2% were survived
 - from total male passengers 577 only 109 i.e 18.8% were survived
3. **Survival rate based on Pclass:**
 - from total class-1 passengers 216 only 136 i.e 62.9% were survived
 - from total class-2 passengers 184 only 87 i.e 47.2% were survived
 - from total class-3 passengers 491 only 119 i.e 24.2% were survived
4. **Survival rate based on AgeGroup:**
 - from total child people 69 only 40 i.e 57.9% were survived
 - from total teen people 95 only 39 i.e 41% were survived
 - from total adult people 422 only 139 i.e 32.9% were survived
 - from total middle age people 202 only 86 i.e 42.5% were survived
 - from total senior people 103 only 38 i.e 36.8% were survived

```
In [77]: # Defining subplots figure 2
fig2, ax2 = plt.subplots(1,2,figsize=[10,5])
```

```
In [78]: # Age Distribution
ax2[0].hist(df.Age,bins=30,color='orange',edgecolor='black')
ax2[0].set_title('Age Distribution')
ax2[0].set_xlabel('Age (0-80)')
ax2[0].set_ylabel('Number of persons')
```

```
# Fare vs Age plot
ax2[1].scatter(df.Age,df.Fare,c=df.Survived,cmap='coolwarm')
ax2[1].set_title('Fare vs Age (Survived=red)')
ax2[1].set_xlabel('Age')
ax2[1].set_ylabel('Fare')

plt.show()
fig2.savefig("Age_Fare.png", dpi=300)
```



- INSIGHTS:
 1. **Age Distribution:**
 - there are more passengers in age group between 20-40
 2. **Fare vs Age:**
 - red dots denote survived people and who paid high fare i.e 500+ was survived

Advanced Visualization

1. Gender based Visualization
2. Pclass based Visualiation
3. Age Group based Visualization

Gender based visualization

In [79]:

```
# Survival rate (pie), survived people & died people (bar + pie)
fig3, ax3 = plt.subplots(2,2,figsize=[10,10])
```

In [80]:

```
xpos4 = np.arange(len(gender_survived))
ax3[0,0].bar(xpos4-0.2,gender_survived.values,label='Survived',width=0.4,color='green')
ax3[0,0].bar(xpos4+0.2,gender_died.values,label='Died',width=0.4,color='orange')
ax3[0,0].set_xticks(xpos4,gender_survived.index)
ax3[0,0].set_title('Survived and died people based on gender')
ax3[0,0].set_xlabel('Gender')
ax3[0,0].set_ylabel('Number of people')
ax3[0,0].legend(labels=['Survived','Died'])

ax3[0,1].pie(gender_survival,autopct='%0.1f%%',labels=gender_survival.index,colors=['lightcoral','crimson'])
ax3[0,1].set_title('Survival Rate based on gender')

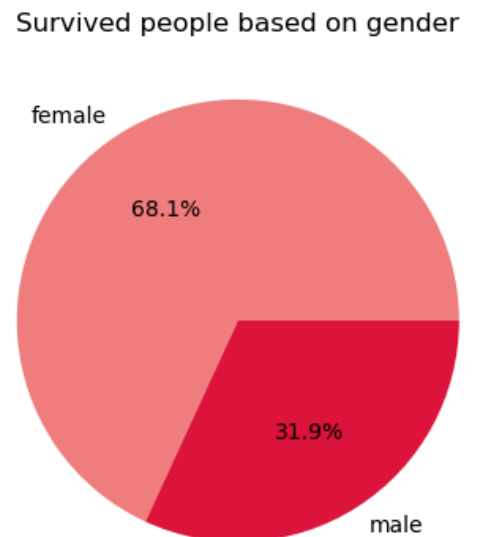
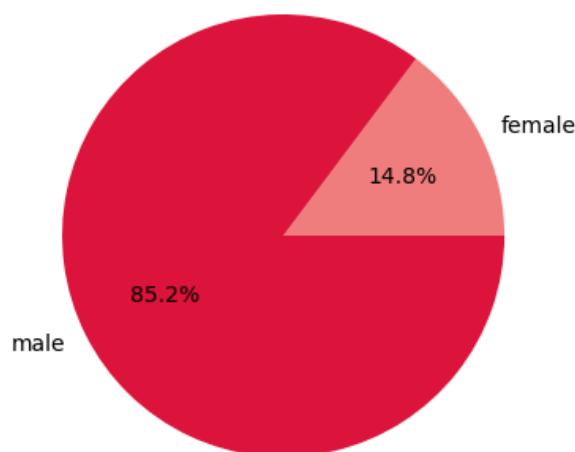
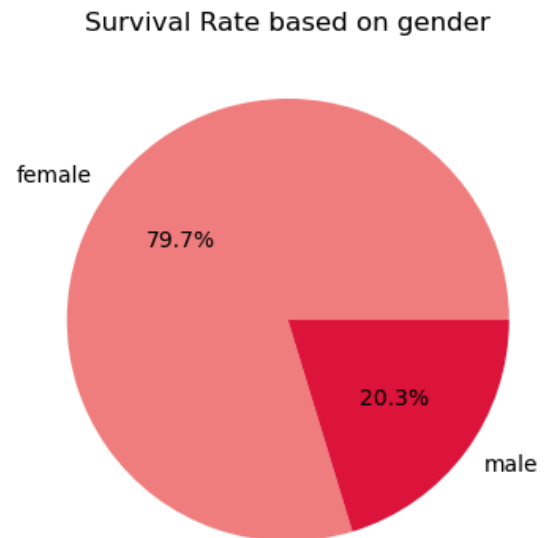
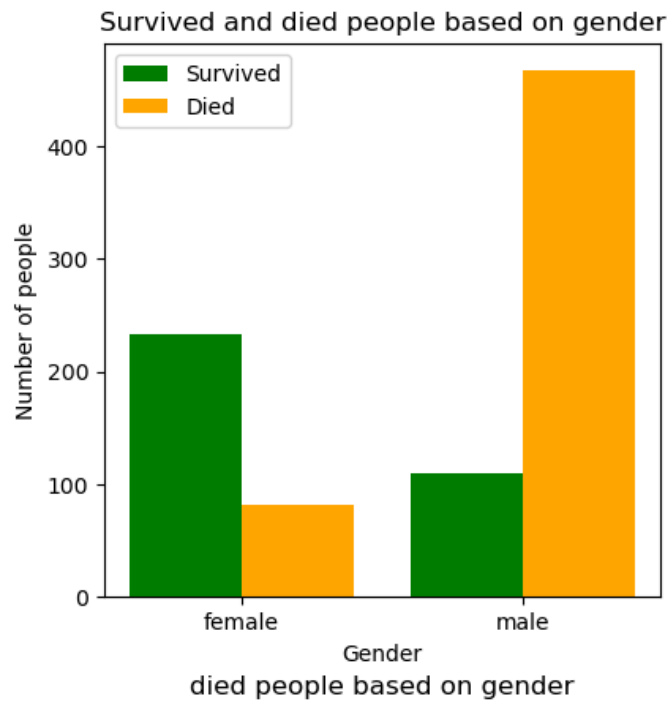
ax3[1,0].pie(gender_died,autopct='%0.1f%%',labels=gender_died.index,colors=['lightcoral','crimson'])
ax3[1,0].set_title('died people based on gender')

ax3[1,1].pie(gender_survived,autopct='%0.1f%%',labels=gender_survived.index,colors=['lightcoral','crimson'])
```

```
ax3[1,1].set_title('Survived people based on gender')
```

```
plt.show()
```

```
fig3.savefig("Gender_based.png", dpi=300)
```



• INSIGHTS:

1. Survived and died people based on gender:

- in total female passengers **314** 81 are died and **233** are survived
- in total male passengers **577** 468 are died and **109** are survived

2. Survival rate based on gender:

- female passengers survival rate is **74.2%**
- male passengers survival rate is **18.8%**

3. Died people based on gender:

- died female passengers are **14.8%** form total died passengers
- died male passengers are **85.2%** from total died passengers

4. Survived people based on gender:

- survived female passengers are **68.1%** from total survived passengers
- survived male passengers are **31.9%** from total survived passengers

Pclass based visualization

```
In [81]: # Survival rate (pie), survived people & died people (bar + pie)
fig4, ax4 = plt.subplots(2,2,figsize=[10,10])
```

```
In [82]: xpos5 = np.arange(len(Pclass_survived))
ax4[0,0].bar(xpos5-0.2,Pclass_survived.values,label='Survived',width=0.4,color='green')
ax4[0,0].bar(xpos5+0.2,Pclass_died.values,label='Died',width=0.4,color='orange')
```



```

ax4[0,0].set_xticks(xpos5,Pclass_survived.index)
ax4[0,0].set_title('Survived and died people based on passenger class')
ax4[0,0].set_xlabel('Passenger class')
ax4[0,0].set_ylabel('Number of people')
ax4[0,0].legend(labels=['Survived','Died'])

classes = ['Class-1','Class-2','Class-3']

ax4[0,1].pie(Pclass_survival,autopct='%0.1f%%',labels=classes,colors=['lightcoral','crimson','hotpink'])
ax4[0,1].set_title('Survival Rate based on passenger class')

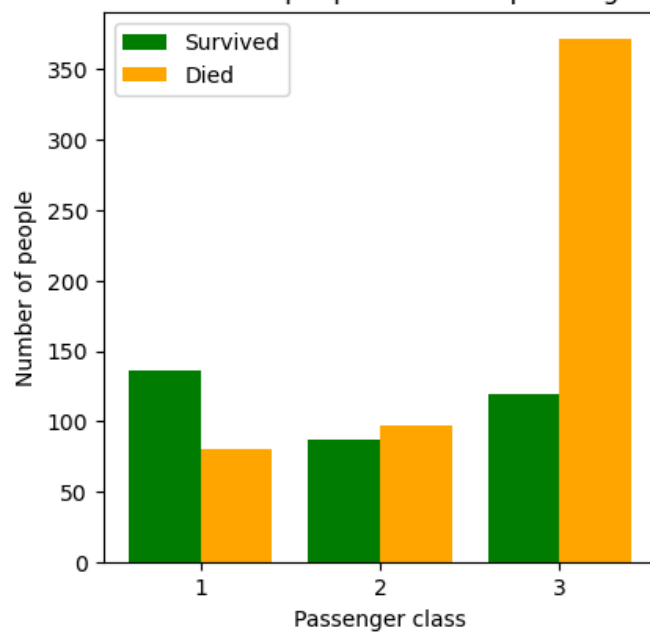
ax4[1,0].pie(Pclass_died,autopct='%0.1f%%',labels=classes,colors=['lightcoral','crimson','hotpink'])
ax4[1,0].set_title('died people based on passenger class')

ax4[1,1].pie(Pclass_survived,autopct='%0.1f%%',labels=classes,colors=['lightcoral','crimson','hotpink'])
ax4[1,1].set_title('Survived people based on passenger class')

plt.show()
fig4.savefig("Pclass_based.png", dpi=300)

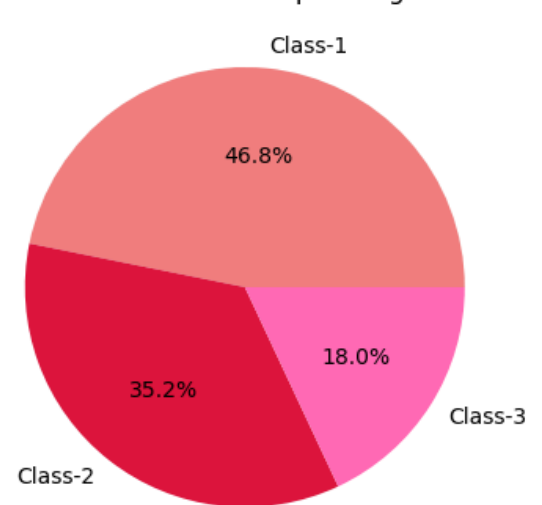
```

Survived and died people based on passenger class

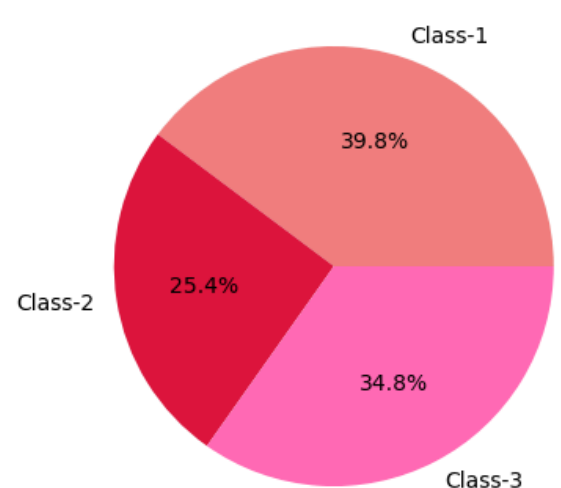
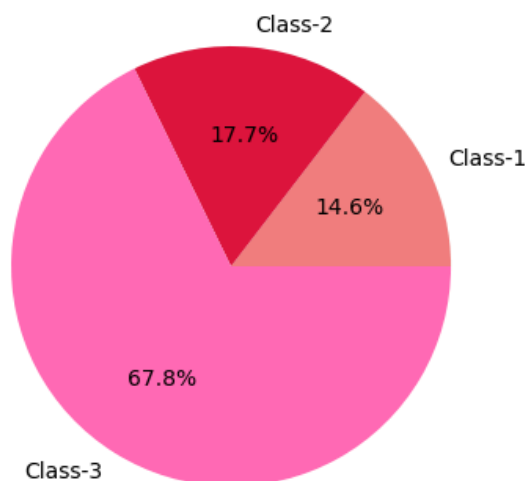


died people based on passenger class

Survival Rate based on passenger class



Survived people based on passenger class



- INSIGHTS:

- Survived and died people based on passenger class:

- in total class-1 passengers 216 80 are died and 136 are survived
- in total class-2 passengers 184 97 are died and 87 are survived
- in total class-3 passengers 491 372 are died and 119 are survived

- Survival rate based on passenger class:

- class-1 passengers survival rate is 46.8%
- class-2 passengers survival rate is 35.2%
- class-3 passengers survival rate is 18.0%

- Died people based on passenger class:

- died class-1 passengers are **14.6%** from total died passengers
 - died class-2 passengers are **17.7%** from total died passengers
 - died class-3 passengers are **67.8%** from total died passengers
4. **Survived people based on gender:**
- survived class-1 passengers are **39.8%** from total survived passengers
 - survived class-2 passengers are **25.4%** from total survived passengers
 - survived class-3 passengers are **34.8%** from total survived passengers

Age Group based Visualization

```
In [83]: # Survival rate (pie), survived people & died people (bar + pie)
fig5, ax5 = plt.subplots(2,2,figsize=[10,10])
```

```
In [84]: xpos6 = np.arange(len(age_group_survived))
ax5[0,0].bar(xpos6-0.2,age_group_survived.values,label='Survived',width=0.4,color='green')
ax5[0,0].bar(xpos6+0.2,age_group_died.values,label='Died',width=0.4,color='orange')
ax5[0,0].set_xticks(xpos6,age_group_survived.index)
ax5[0,0].set_title('Survived and died people based on age group')
ax5[0,0].set_xlabel('Age group')
ax5[0,0].set_ylabel('Number of people')
ax5[0,0].legend(labels=['Survived','Died'])

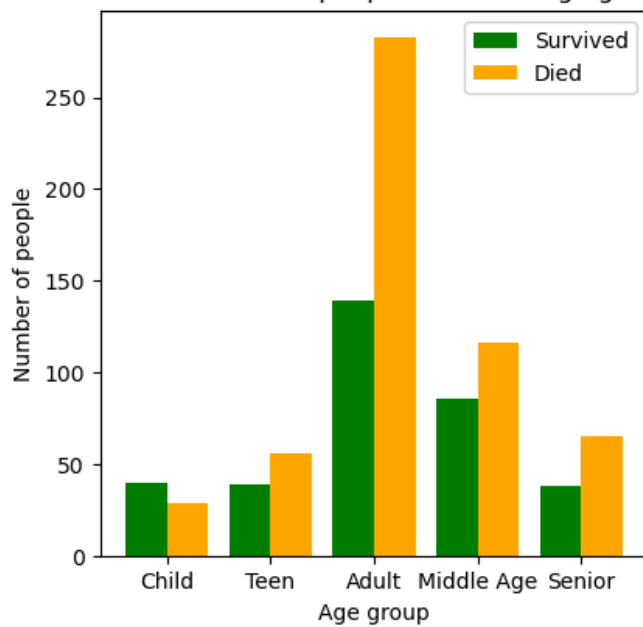
ax5[0,1].pie(age_group_survival,autopct='%0.1f%',labels=age_group_survival.index)
ax5[0,1].set_title('Survival Rate based on age group')

ax5[1,0].pie(age_group_died,autopct='%0.1f%',labels=age_group_died.index)
ax5[1,0].set_title('died people based on age group')

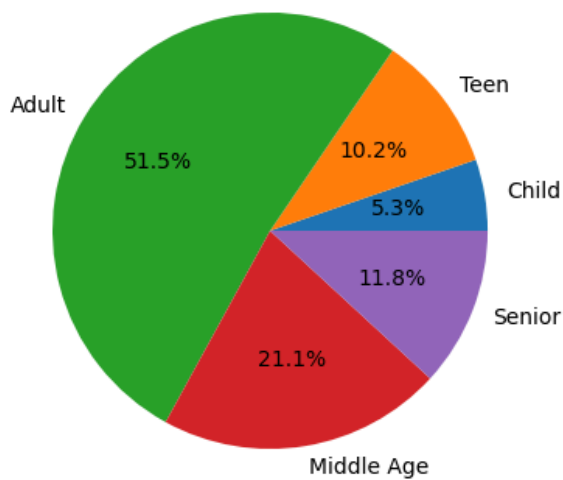
ax5[1,1].pie(age_group_survived,autopct='%0.1f%',labels=age_group_survived.index)
ax5[1,1].set_title('Survived people based on age group')

plt.show()
fig5.savefig("AgeGroup_based.png", dpi=300)
```

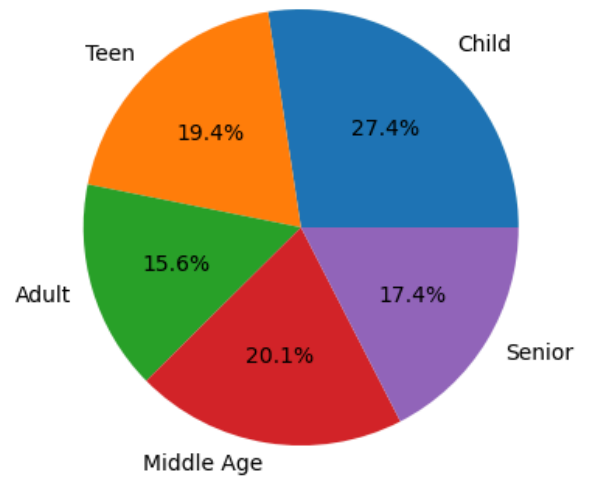
Survived and died people based on age group



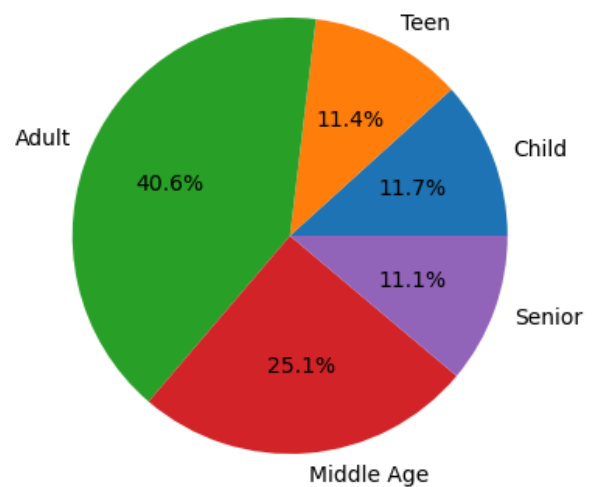
died people based on age group



Survival Rate based on age group



Survived people based on age group



• INSIGHTS:

1. Survived and died people based on age group:

- in total child passengers 69 29 are died and 40 are survived
- in total teen passengers 95 56 are died and 39 are survived
- in total adult passengers 422 283 are died and 139 are survived
- in total middle age passengers 202 116 are died and 86 are survived
- in total senior passengers 103 65 are died and 119 are survived

2. Survival rate based on age group:

- child passengers survival rate is 27.4%
- teen passengers survival rate is 19.4%
- adult passengers survival rate is 15.6%
- middle age passengers survival rate is 20.1%
- senior passengers survival rate is 17.4%

3. Died people based on age group:

- died child passengers are 5.3% from total died passengers
- died teen passengers are 10.2% from total died passengers
- died adult passengers are 51.5% from total died passengers
- died middle age passengers are 21.1% from total died passengers
- died senior passengers are 11.8% from total died passengers

4. Survived people based age group:

- survived child passengers are 11.7% from total survived passengers
- survived teen passengers are 11.4% from total survived passengers
- survived adult passengers are 40.6% from total survived passengers
- survived middle age passengers are 25.1% from total survived passengers
- survived senior passengers are 11.1% from total survived passengers

Conclusion

- female and child passengers have more survival rate.
- most of the male passengers were died
- most of the middle age, adult passengers died
- most of the class-3 people died

